Think — ratio in context

Scales on maps and plans, best value in a shop, and ratios that describe the world.

LESSON 54 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 12 APPLICATION

LESSON CLOCK

12' | 12' | 10' | 6'

Scales	12 min
Best value	12 min
Ratios in the world	10 min
Is the model reasonable?	6 min

LEARNING OBJECTIVES

- Read and use the scale of a map or plan.
- Compare prices by unit cost to find the better value.
- Interpret a ratio given in a real context.
- Judge whether a proportional model is reasonable.

TEXTBOOK

National	pp. 151–153
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.6

01 Explanation

Scales

A scale is a ratio: *1 : 100 000* means one centimetre on the map is *100 000* centimetres — one kilometre — on the ground.

SCALE	*1 cm* REPRESENTS	*5 cm* REPRESENTS
1 : 100 000	*1 km*	*5 km*
1 : 25 000	*250 m*	*1.25 km*
1 : 50	*50 cm*	*2.5 m*
1 : 200	*2 m*	*10 m*

CONVERT THE UNITS AT THE END, NOT IN THE MIDDLE

A scale of *1 : 25 000* turns *3 cm* into *75 000 cm*; only then divide by *100 000* to get *0.75 km*. Converting halfway through is where the factors of ten go missing.

Best value

Compare the cost of one unit — or the amount for one sum — and the better buy is obvious.

OFFER	COST	PER UNIT	VERDICT
3 pens	*12 000*	*4000*	—
5 pens	*18 000*	*3600*	better
500 g of rice	*9 000*	*18 a gram*	—
2 kg of rice	*32 000*	*16 a gram*	better

THE BIGGER PACK USUALLY WINS — BUT NOT ALWAYS

Working out the unit cost takes ten seconds and settles it. Shops rely on customers assuming rather than dividing.

Ratios in the world

CONTEXT	RATIO	WHAT IT TELLS YOU
a map	$1 : 50\,000$	how much the ground is shrunk
concrete	$1 : 2 : 4$	the recipe for a strong mix
a school	$1 : 20$ teachers to pupils	the size of a class
a photograph enlarged	$1 : 3$	every length trebled
gears on a bicycle	$2 : 5$	turns of the wheel per pedal turn

A RATIO COMPRESSES A WHOLE SENTENCE

“For every teacher there are twenty pupils” is one line of English; $1 : 20$ is four characters, and it can be calculated with.

Is the model reasonable?

CLAIM	PROPORTIONAL?	COMMENT
3 pens cost three times one pen	yes	if the price is fixed
100 pens cost a hundred times one	usually not	bulk discounts
2 painters take half the time	roughly	up to a point
50 painters take a fiftieth of the time	no	they cannot all reach the wall
a child twice as old is twice as tall	no	growth is not proportional

EVERY MODEL HAS A RANGE WHERE IT WORKS

Proportion is an excellent description of prices and speeds, and a poor one of growth and crowded workplaces. Saying where a model stops working is part of using it well.

02 Worked examples

EXAMPLE 1 **A map has scale $1 : 25\,000$. Two towns are 8 cm apart on it. How far apart are they really?**

1. $8 \cdot 25\,000 = 200\,000$ cm.

   Multiply by the scale.

2. $200\,000 \div 100\,000 = 2$.

   Convert at the end.

3. 2 km.

ANSWER

2 km

EXAMPLE 2 **Which is better value: *3* pens for *12 000* or *5* for *18 000*?**

1. $12\ 000 \div 3 = 4000$ each.
2. $18\ 000 \div 5 = 3600$ each.
3. The pack of five is better value.

ANSWER

5 for *18 000*

EXAMPLE 3 **A plan of a room has scale *1 : 50*. The room is *4 m* long. How long is it on the plan?**

1. $4\ m = 400\ cm$.
2. $400 \div 50 = 8$.
   Divide, because the plan is smaller.
3. *8* cm.

ANSWER

8 cm

03 Interactive model

Bring a real map and a supermarket receipt; both are ratio problems, and neither looks like a textbook exercise.

Scales and best value

At $1 : 100\ 000$, $1\ cm$ is:

- $100\ m$
- $1\ km$
- $10\ km$
- $100\ km$

At $1 : 25\ 000$, $8\ cm$ is:

- 2 km
- 20 km
- 200 m
- 25 km

At $1 : 50$, a $4\ m$ room is drawn:

- 8 cm
- 80 cm
- 2 cm
- 20 cm

3 pens for *12 000* cost each:

- *3000*
- *4000*
- *6000*
- *36 000*

5 for *18 000* cost each:

- *3600*
- *4000*
- *3000*
- *9000*

So the better value is:

- *3* for *12 000*
- *5* for *18 000*
- the same
- cannot say

50 painters taking a fiftieth of the time is:

- reasonable
- unreasonable
- certain
- proportional

A scale of $1 : 200$ means $1\ cm$ is:

- 200 cm
- 2 m
- both of these
- 200 m

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Scale	Masshtab	Масштаб
Map	Xarita	Карта
Plan	Reja	План
Unit cost	Birlik narxi	Цена за единицу
Best value	Eng foydali	Наиболее выгодно
Real distance	Haqiqiy masofa	Реальное расстояние
Model	Model	Модель
Reasonable	Maqbul	Разумный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A map scale is a:

- fraction
- ratio
- percentage
- proportion of area

To go from map to ground you:

- divide by the scale
- multiply by the scale
- add it
- square it

To go from ground to map you:

- divide by the scale
- multiply by the scale
- subtract it
- do nothing

Best value is found by comparing:

- total prices
- unit costs
- pack sizes
- shops

At $1 : 25\,000$, $4\ cm$ is:

- 1 km
- 10 km
- 100 m
- 25 km

Proportion describes growth in height:

- well
- badly
- exactly
- always

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. At $1 : 100\ 000$, $1\ cm$ represents

ANSWER 1 km

2. At $1 : 25\ 000$, $1\ cm$ represents

ANSWER 250 m

3. At $1 : 50$, $1\ cm$ represents

ANSWER 50 cm

4. At $1 : 100\ 000$, $5\ cm$ represents

ANSWER 5 km

5. 3 pens for $12\ 000$: one pen

ANSWER 4000

6. 5 pens for $18\ 000$: one pen

ANSWER 3600

7. Which is better value?

ANSWER The pack of five

Medium · the standard question

7 PROBLEMS

1. At $1 : 25\ 000$, $8\ cm$ represents

ANSWER 2 km

2. At $1 : 50$, a $4\ m$ room is drawn

ANSWER 8 cm

3. 500 g for $9\ 000$ or 2 kg for $32\ 000$

ANSWER The 2 kg pack

4. A photo enlarged $1 : 3$: a $5\ cm$ side becomes

ANSWER 15 cm

5. At $1 : 200$, $7\ cm$ represents

ANSWER 14 m

6. A $12\ m$ wall at $1 : 200$

ANSWER 6 cm

7. Teachers to pupils $1 : 20$ with 300 pupils

ANSWER 15 teachers

Hard · stretch and reason

7 PROBLEMS

1. At $1 : 50\ 000$, a real distance of 3.5 km is drawn

   ANSWER 7 cm

2. A map at $1 : 25\ 000$ and one at $1 : 100\ 000$: which shows more detail?

   ANSWER The $1 : 25\ 000$

3. 750 g for $15\ 000$ or 1.2 kg for $22\ 800$

   ANSWER The 1.2 kg pack, at 19 a gram

4. A model car at $1 : 43$ is $10\ cm$ long: the real car

   ANSWER 4.3 m

5. Why is “100 pens cost 100 times one” often false?

   ANSWER Bulk prices are usually discounted

6. Gears $2 : 5$: wheel turns for 30 pedal turns

   ANSWER 75

7. At what scale is $1\ cm$ equal to $1\ m$?

   ANSWER $1 : 100$

07 Homework

Homework — 5 tasks

Convert units only at the end of the calculation.

1. At a scale of $1 : 25\ 000$, find the real distance for $6\ cm$.
2. At a scale of $1 : 50$, find the plan length of a $7\ m$ wall.
3. Which is better value: 4 notebooks for $20\ 000$ or 7 for $33\ 600$?
4. A map shows two towns $4.5\ cm$ apart at $1 : 200\ 000$. Find the real distance.
5. Give one situation where proportion describes the world well and one where it does not.